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EDUCATION AND SCHOLASTIC ACHIEVEMENTS

Program	Institute	% / CGPA	Year
B. Tech. (Ocean Engg. & IDDD Robotics)	Indian Institute of Technology, Madras	8.34	2027
STD XII	M.P. Board	86%	2022
STD X	C.B.S.E.	83.8%	2020

- Qualified for JEE Advance and secured a position in top 0.9% among 1.1 million participants conducted by IIT Bombay.
- Ranked among the top 0.4 percentile out of 9 lakh candidates nationwide in the competitive JEE Mains 2022 examination.
- Achieved a maximum rating of 1391 at Codeforces by constantly participating in competitive programming contests.

PROFESSIONAL EXPERIENCE

Internship at Deutsche Bank Analyst role
(May 26 – July 26)

- Developed a **vulnerability remediation** pipeline using **GitHub Actions** with Scan, Triage, Remediation, and Validation agents to automate dependency **vulnerability detection**, **prioritization**, **updation**, **build check**, and **PR generation.**
- Developed an **automated validation pipeline** that verifies dependency updates by **modifying package manager files**, **executing CI builds**, **reverting failed upgrades**, and **preserving only successful, buildable changes**.
- Automated dependency remediation using **Renovate Bot, deployed** on server instance for **Bitbucket repositories** using **Artifactory** and **TeamCity** to **validate builds** of dependency updates and **create pull requests for successful builds**.
- Developed Python tool to **parse Maven dependency trees** and integrate with **OSV, NVD** to **identify vulnerabilities**, retrieve CVEs, severity scores, and fixed versions for both **direct and transitive dependencies.**
- Received a **Pre-Placement Offer (PPO)** for delivering scalable automation solutions that streamlined enterprise dependency vulnerability management and **reduced manual remediation effort** by **40%.**

Cybersecurity Internship at IIT Madras
(May 24 – Aug 24)

R&D internship under Prof. NS Narayanaswamy

- Explored **Phishing** techniques using **Gophish** and **Evilginx** to simulate real-world scenarios and **credential harvesting**.
- Developed **PoC payloads** to demonstrate **initial access techniques** by delivering payloads on target systems in a sandbox.

PERSONAL PROJECTS

Latent Diffusion Model (LDM)

- Developed LDM from scratch using **PyTorch**, implemented **UNet** for noise prediction, **VAE** for latent space representation.
- Leveraged **Hugging Face** library to **tokenize 10,000-words** integrated **CLIP** for **prompt-based image generation.**
- Enabled controlled image generation by embedding prompts into **cross-attention** and aligning text with visuals precisely

Custom CNN Model for Card detection

- Achieved **91% accuracy** on custom **CNN model** by developing a 4-layer architecture for **reliable detection of rank & suit**
- **Optimized feed-forward time** by incorporating **batch normalization** layers and using OpenCV for live camera input feed
- **Reduced overfitting** by Applied **data augmentation** techniques, **L2 regularization,** and **dropout** for better generalization.

PID Controller

- Achieved precise **motion control** for **autonomous vessels** using **PID closed-loop controller** within the **ROS2** framework.
- Achieved **real-time position tracking** by utilizing **UWB** sensors and maneuvering equations to simulate vessel dynamics.
- Computed mass and Inertial matrices for accurate **motion prediction** and implemented them to enhance system stability.

Ace Bot

- Engineered a **robotic manipulator** using **CNN-based card detection**, enabling automated gameplay with human players.
- Engineered **forward** and **inverse kinematics** using **Jacobian inversion** for smooth and stable manipulator motion control.
- Optimized weight distribution and structural design by addressing challenges from motor weight, ensuring stable operation

POSITIONS OF RESPONSIBILITY

Envisage Core (Shaastra, CFI)
(Apr 25 – Apr 26)

- Led **Team Envisage**, a multidisciplinary team comprising **Tech Ambience**, **Shows**, **GameDev**, and **XR** as verticals
- Represented a **70+ member team** across the **co-curricular sphere** and Oversaw **end-to-end operations of the team**
- Represented a 1**000+ member** organizing body as **Shaastra core** delivering key contributions across **India's largest student-run tech fest.**

Envisage Manager
(Mar 24 – Mar 25)

- Managed a team of **20+ members**, overseeing project execution, logistics and coordination for technical deliverables.
- Worked on **Robotic manipulators** and implemented inverse kinematics using Jacobian inversion and developed custom CNN models

Envisage Coordinator
(Jul 23– Mar 24)

- Coordinated with the team to develop Swift Canvas, advancing the previous year's **2D pen plotter** project.
- Showcased the project at **CFI Open House** 2024, **CFI Research Conclave** 2023, and **Shaastra** 2024, engaged over 9300.

COURSE WORK & SKILLS

OB3214: Deep learning for Computer Vision	EE6180: Advanced Topics in Artificial Intelligence
ID6040: Introduction to Robotics	MA2040: Probability Statistics and Stochastic Process
OB2202: Programming, Data Structures and Algorithms using Python	CS6847: Cloud Computing
ED5215: Introduction to motion planning	OE5011: Marine Robotics
EE6347: Devices and technologies for AI and neuromorphic computing	EE6412: Optimal Control

Libraries: Pytorch, Numpy, Pandas, Hugging Face, Gymnasium, Streamlit, Mediapipe, OpenCV, Scikit-learn, Flask
Technologies: Python, C, C++, Bash, Shell, Node, MatLab, SQL, ROS2, Linux, Arduino
Software: Git, GitHub, Docker, GitHub Actions, Helm, K8s, Fusion 360, Unity, Jupyter Notebook, Colab, Kaggle, BitBucket, Artifactory, TeamCity

AWARDS AND CERTIFICATIONS

- Received Achiever's award for securing **second position in Class XII** amongst 200+ students.
- Awarded for **Highest Overall Distinction** among 120+ students in **Class X** earning school-wide recognition for best overall performance.